Tawanda Chiteshe

Full-stack & Systems Engineer

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SUMMARY

Self-taught engineer, now studying software engineering, who has shipped production systems for clients since 2020. I build multi-tenant SaaS backends, offline-first desktop apps, GIS APIs and auth-heavy web portals in Go, Rust, TypeScript and Kotlin. Comfortable down to the metal: I have written game engines, a CPU emulator and an audio synthesizer.

TECHNICAL SKILLS

Languages	Go, Rust, TypeScript, Kotlin, Java, Python, C++, SQL
Backend	NestJS, Quarkus, Spring Boot, FastAPI, Ktor, GraphQL, gRPC, REST + OpenAPI, SQLAlchemy, Celery
Frontend & apps	React, Next.js, Vite, Tailwind CSS, Tauri 2, Compose Multiplatform
Data	PostgreSQL, PostGIS, SQLite, Prisma, Flyway, Valkey/Redis, MinIO/S3
Infra & auth	Docker, Kubernetes (k3s), Flux CD, CloudNativePG, OpenBao, Podman, Traefik, nginx, Keycloak, Authentik, OIDC/JWT, FreeRADIUS, Linux

EXPERIENCE

Software Developer — Nerd Zero 2025 – present

- **listnun, newsletter platform as a service (Go, React):** building a multi-tenant SaaS on a shared Listmonk fork. Moved from one pod per tenant to one shared fork scoped by Host header, so adding a tenant is a database row, not a deployment.
- Integrated Postmark's Account API so every tenant gets its own sending server; built custom domains, metered billing, invoices and a worker that suspends lapsed accounts.
- **Homwe POS, offline-first point of sale (Rust, Tauri 2, React):** Rust owns the GraphQL client, SQLite, sale arithmetic and ESC/POS receipts. Sales upload in idempotent batches with retries, so no sale is lost when the connection drops.
- Added LAN printer discovery, USB printing on Linux and Windows, and a customer display on a second monitor.
- **Infrastructure (Kubernetes):** deployed listnun to the company's k3s clusters on Hetzner Cloud through GitOps (Flux CD), managed the shared PostgreSQL cluster (CloudNativePG) and set up OpenBao for secrets management.

Software Developer — Vuleka 2025 – 2026

- Built the NestJS 11 + Prisma API for a distribution platform: merchants, inventory, ordering and deliveries.
- Added Keycloak role-based auth, MinIO file storage and WhatsApp order notifications through WATI; worked on the Next.js frontend.

Software Developer — Runhare 2023 – present

- Building an interactive e-learning platform with separate tutor and learner dashboards.
- Built the Quarkus (Java) API: JWT auth, PostgreSQL with Flyway migrations, S3 file storage, email and scheduled jobs, served with a Next.js frontend.

Software Developer — Kgopelang Health 2023

- Built BANS (Business Admin Notification System): a NestJS + PostGIS API and Next.js app for managing patient cases, nurse bookings, rapid medical tests and insurer integrations.
- Secured it with Keycloak and ran it in Docker.

Freelance 2025 – 2026

- **Agritech GIS platform:** Go API on PostGIS with Keycloak auth, plus a FastAPI + Celery service that computes NDVI from satellite rasters.
- **Wi-Fi hotspot billing:** NestJS + FreeRADIUS server and Next.js captive portal for paid vouchers (live demo).

Software Developer — Elula Online Dec 2020 – Jan 2021

- Built a Python gRPC service using SQLAlchemy for database access.

SELECTED PROJECTS

- **school-portal** (React, FastAPI, PostgreSQL): college onboarding portal that reads ZIMSEC certificates with OCR, validates Zimbabwe national IDs and includes an AI assistant limited to each student's own data.
- **coinage** (Kotlin Multiplatform, Compose, SQLDelight, Ktor): local-first finance tracker from one codebase on Android, iOS, desktop and web.
- **mbirasyth** (Rust, DSP): sample-free Shona mbira synthesizer and procedural composer for games.
- **crust-6502-emulator** (Rust): MOS 6502 CPU emulator.
- **cuboid** (C++, OpenGL, Lua, EnTT): game engine and editor designed for low-end hardware.

EDUCATION

Diploma in Software Engineering — TelOne Centre for Learning (TCFL) 2026 – present

Studying alongside full-time work at Nerd Zero.